Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids: A Latency–Robustness Trade-off Analysis

Jorge Luis Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez
Department of Electrical and Electronics Engineering
Universidad de los Andes
Bogotá, Colombia
jl.mayorga236@uniandes.edu.co, ya.africano1030@uniandes.edu.co,
fjimenez@uniandes.edu.co

***Abstract*—Frequency estimation for relay and monitoring functions is more demanding in inverter-based resource (IBR) grids than in classical synchronous systems because phase jumps, harmonics, ringdown, and out-of-band components may occur in the same event. This paper benchmarks 16 estimators over 34 scenarios with 60 Monte Carlo runs per estimator–scenario pair, covering IEEE-style ramps, magnitude steps, modulation, phase jumps, out-of-band interference, and IBR-specific composite stress. No single estimator dominates all disturbance families. The second-order generalized-integrator PLL (SOGI-PLL) leads ramps, the extended Kalman filter (EKF) leads magnitude steps and out-of-band cases, and the robust adaptive EKF (RA-EKF) leads modulation. The IBR multi-event case splits RMSE and trip-risk leadership between the phase-locked loop (PLL) and the interpolated DFT (IpDFT). The hardest cases are the IBR multi-event sequence, 60-degree phase jumps, and noisy ringdown scenarios. These results show that isolated compliance tests miss disturbance dependence that matters for protection design.**

***Index Terms*—Frequency Estimation, IBR Grids, Low-Inertia Systems, Monte Carlo Benchmark, Relay Protection.**

I. Introduction

Frequency estimation sits on the critical path of relaying, under-frequency load shedding, disturbance monitoring, and converter supervision in low-inertia grids. In inverter-based resource (IBR) systems, the problem is not limited to slow electromechanical drift. Harmonics, abrupt phase jumps, out-of-band components, and short ringdown segments can arrive on the same timescale as the protection decision itself [1]–[3]. Recent IBR-focused studies also show that converter control interactions and mixed disturbances can reshape the observable frequency response itself [4], [5]. An estimator that performs well in a textbook sinusoidal test can still generate large transient error or non-negligible trip-risk when these effects overlap.

The estimator families used in practice already reveal a tension between accuracy, latency, and disturbance tolerance. Windowed spectral methods reject steady harmonic content but pay observation delay [6], [7]. PLL variants are light enough for embedded deployment but can slip under abrupt phase discontinuities [8], [9]. Kalman-filter variants can track fast transients when the state model remains valid [10], [11]. Reduced-order and data-driven methods offer another point on the cost-accuracy curve, but their behavior under relay-oriented composite stress is even less standardized [12], [13]. What remains unclear is how the ranking of these families changes once the benchmark moves beyond isolated standard tests.

This paper treats that question as the main result. Rather than forcing a single winner across all conditions, we ask three practical questions: which estimators lead the standard disturbance families, which IBR stress cases reorder the ranking, and which methods remain attractive once CPU cost and trip-risk are considered alongside RMSE.

The benchmark analyzed here covers 16 estimators and a broad scenario set that mixes IEEE- and NERC-style disturbances with IBR-specific harmonics, ringdown, and multi-event stress. The dominant estimator changes with the disturbance family: the second-order generalized-integrator PLL (SOGI-PLL) leads ramps, the extended Kalman filter (EKF) leads magnitude steps and out-of-band interference, the robust adaptive EKF (RA-EKF) leads modulation, and the IBR multi-event case splits RMSE and trip-risk leadership between the phase-locked loop (PLL) and interpolated DFT (IpDFT). That scenario dependence, rather than any single ranking, is the main message of the paper.

A. Related Work

Standards and many published benchmark studies still center on isolated disturbance classes. IEEE C37.118.1 and IEC/IEEE 60255-118-1 define dynamic tests around ramps, magnitude steps, modulation, and a limited set of phase events [14], [15]. Ting et al. extend that logic to low-inertia frequency measurement methods and show clear trade-offs between tracking speed and disturbance rejection even under a carefully chosen set of simulated events [13]. Wright et al. further show that phase steps can produce false ROCOF signatures, which is precisely the kind of measurement ambiguity that simple compliance views do not resolve [9].

Recent IBR-focused studies widen rather than close this gap. Low-inertia surveys and system studies show that converter penetration changes both the spatial distribution and the time scale of frequency dynamics [1]–[3]. Related benchmark proposals for hybrid islanded or grid-forming systems emphasize interaction-driven behavior, but they do not provide a broad estimator comparison over standard PMU tests, composite IBR stress, trip-risk, and compute cost in one frame [4], [5]. That is the limitation this paper addresses. A method that performs well on a clean standard test may fail under conditions that the test never probes.

The estimator families themselves are not the missing piece. Window-based methods trade harmonic rejection against observation delay [6], [7]. PLL variants are cheap enough for embedded deployment but can lose phase alignment under abrupt discontinuities [8], [9]. Kalman-filter variants can improve transient tracking when their state model matches the event [10], [11]. Reduced-order and data-driven methods add further cost-accuracy trade-offs [12]. What is missing is not the estimator families but the comparison protocol: a broad benchmark that keeps scenario family, trip-risk, and compute cost in view at the same time. The contribution of this paper is therefore not a new estimator but a benchmark study that preserves standard families, adds composite IBR stress, and compares accuracy, transient risk, and compute cost within one reporting framework.

II. Methodology

The benchmark compares 16 frequency estimators on a common dynamic test set using waveform generation at 1 MHz and estimator outputs at 10 kHz. The estimator pool spans zero-crossing, windowed spectral, PLL-based, Kalman-filter, adaptive, and reduced-order designs: ZCD, IpDFT, TFT, RLS, PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL, LKF2, EKF, UKF, RA-EKF, TKEO, Prony, ESPRIT, and Koopman (RK-DPMU). This mix is broad enough to expose the main trade-offs of interest in protection and monitoring: accuracy, transient risk, settling behavior, and computational burden.

We use standard acronyms throughout. The main ones are the zero-crossing detector (ZCD), interpolated DFT (IpDFT), Taylor-Fourier transform (TFT), phase-locked loop (PLL), and second-order generalized-integrator PLL and FLL (SOGI-PLL and SOGI-FLL). We also use extended Kalman filter (EKF), unscented Kalman filter (UKF), robust adaptive EKF (RA-EKF), and estimation of signal parameters via rotational invariance techniques (ESPRIT).

The study contains 34 benchmark scenarios. Rather than read the benchmark as a flat scenario list, we group the scenarios by disturbance family. Table I summarizes the resulting ten-family structure, and Fig. 1 shows representative ground-truth signals for six of those families.

The family structure reflects distinct physical stress modes. Ramps probe sustained RoCoF tracking. Magnitude steps and modulation probe frequency-amplitude decoupling. Phase jumps probe re-lock behavior and transient overshoot. Out-of-band interference and IBR harmonics probe spectral leakage. Ringdown and multi-event cases combine these effects under changing noise and transient recovery, which makes them the strongest stress tests in the benchmark. Figure 1 illustrates that difference directly: some benchmark families are isolated compliance-style events, whereas others stack abrupt discontinuities, ramps, and oscillatory recovery inside the same record.

TABLE I
Benchmark scope used in this study. Counts refer to benchmark scenarios.

Family	Count	Stress tested
IEEE ramps	9	Sustained RoCoF from 0.25 to 20 Hz/s
Mag. steps	7	1% to 50% magnitude changes
Modulation	3	Nominal, AM, and FM tracking
Phase jumps	3	IEEE 20° and IEEE/NERC 60° re-lock tests
OOB interference	1	Spectral rejection under off-nominal content
Single-tone	1	Deterministic steady-state control case
IBR harmonics	3	Small, medium, and large harmonic stress
IBR ringdown	5	Power-imbalance ringdown with noise sweep
IBR multi-event	1	Composite phase jump, ramp, harmonics, and ringdown
Freq. step	1	Isolated frequency-step control case

Each estimator–scenario pair is tuned via grid search across 100 trials and then evaluated over 60 Monte Carlo runs. The study therefore contains 544 estimator–scenario summaries derived from 32,640 Monte Carlo records. Let $e[k] = \hat{f}[k] - f_{\text{true}}[k]$. We track $\text{RMSE} = \sqrt{N^{-1}\sum_k e[k]^2}$, the maximum peak error $\max_k |e[k]|$, trip-risk duration $T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\}$, settling time, and CPU time per sample. CPU time is interpreted as a relative computational-cost indicator rather than a device latency. For any metric m, the family summary is $\bar{m}_F = |F|^{-1}\sum_{s\in F}\bar{m}_s$, where $\bar{m}_s$ is the scenario mean over the 60 realizations. This prevents dense families from dominating the comparison. Unless stated otherwise, frequency errors are reported in mHz in the tables and discussion.

III. Simulation Results and Benchmarking

A. No Single Estimator Dominates the Standard Scenario Families

The standard disturbance families do not yield a single ranking (Table II). SOGI-PLL gives the lowest mean RMSE on IEEE ramp cases (19.4 mHz). EKF dominates the magnitude-step family (0.388 mHz) and out-of-band interference (13.6 mHz), and RA-EKF leads the modulation family (17.8 mHz). Zero-trip-risk ties are common in ramps, magnitude steps, and modulation, which means those families separate the estimators mainly by accuracy rather than by protection-oriented risk. The margins are often small: the ramp-family gap between SOGI-PLL and Koopman is only 2.7 mHz, while the gap between EKF and RA-EKF in magnitude steps is 1.45 mHz.

The scenario-level winners show the same fragmentation. Across the 34 benchmark scenarios, nine different estimators win at least one scenario on RMSE, and no estimator wins more than 10 of them. Even inside the ramp family, SOGI-PLL wins six of the nine RMSE scenarios while RA-EKF

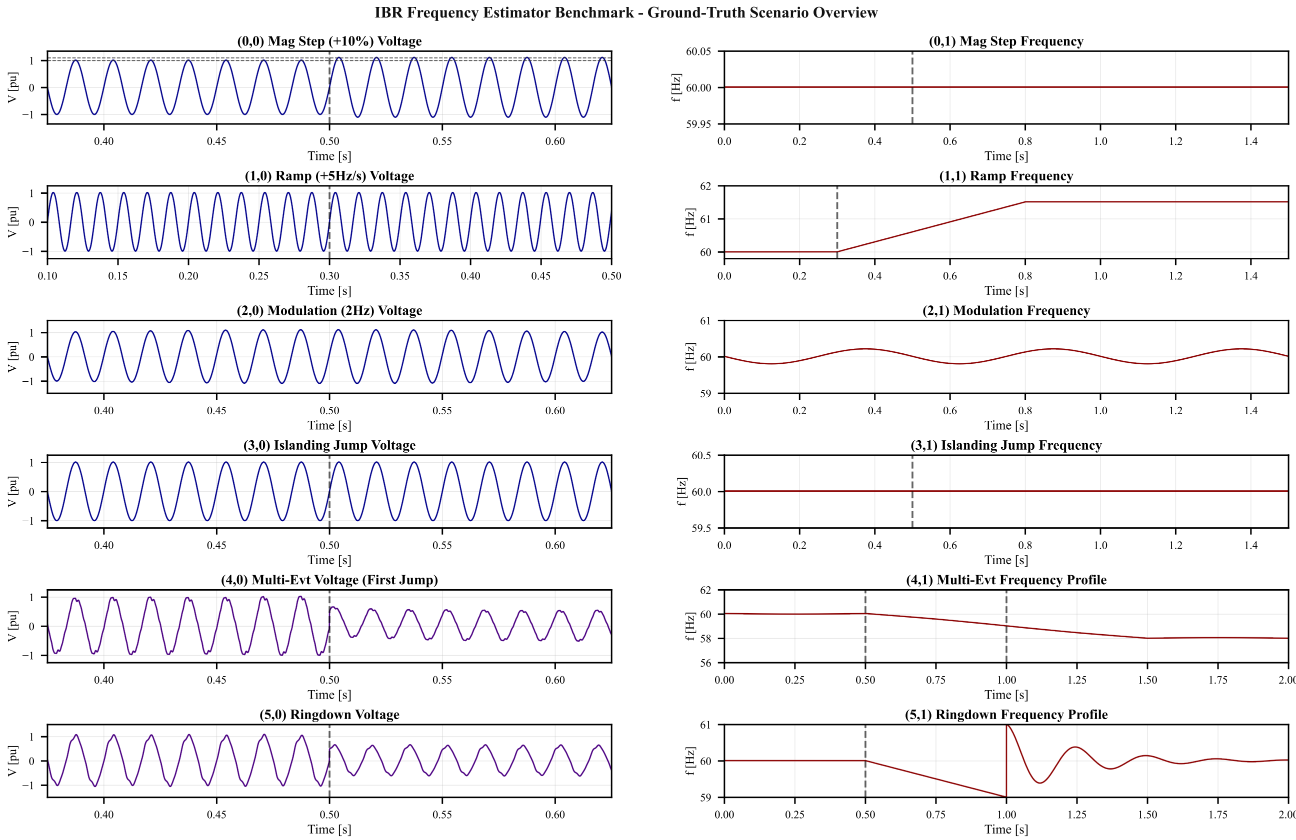

Fig. 1. Representative ground-truth signals in the benchmark. Each row shows a disturbance family with its voltage waveform and true frequency trajectory: (a)–(b) a 10% magnitude step, (c)–(d) a 3 Hz/s ramp, (e)–(f) modulation, (g)–(h) a 60° phase jump, (i)–(j) an IBR multi-event sequence, and (k)–(l) an IBR power-imbalance ringdown. Dashed lines mark disturbance onset. The figure motivates the family-based analysis used in the rest of the paper: the benchmark mixes clean standard tests with composite IBR events that combine several stressors in one record.

TABLE II

CORE DISTURBANCE-FAMILY COMPARISON FOR THE BENCHMARK STUDY. RMSE VALUES ARE REPORTED IN MHZ AND COMPUTED AS FAMILY MEANS OF SCENARIO MEANS. THE UNCERTAINTY TERM IS THE MEAN WITHIN-SCENARIO MONTE CARLO STANDARD DEVIATION FOR THE WINNING ESTIMATOR. Δ IS THE RMSE GAP BETWEEN THE WINNER AND THE RUNNER-UP.

Family	Count	RMSE winner	RMSE [mHz]	Runner-up	Δ [mHz]	Lowest $T_{\mathtt{trip}}$ [s]
IEEE ramps	9	SOGI-PLL	19.4 ± 6.4	Koopman	2.7	0 (multiple)
Mag. steps	7	EKF	0.388 ± 0.143	RA-EKF	1.45	0 (multiple)
Modulation	3	RA-EKF	17.8 ± 21.4	Koopman	13.7	0 (multiple)
Phase jumps	3	SOGI-FLL	174 ± 67.5	PLL	4.03	ZCD: 0.0194
OOB interf.	1	EKF	13.6 ± 11.1	RA-EKF	7.35	EKF, RA-EKF: 0
IBR harmonics	3	EKF	70.5 ± 27.3	TFT	4.13	TFT, Prony: 0.00947
IBR ringdown	5	UKF	313 ± 23.2	EKF	6.67	ESPRIT: 0.0989
IBR multi-event	1	PLL	173.5 ± 66.7	UKF	2.74	IpDFT: 0.0439

takes the other three. The magnitude-step family is the clearest exception: EKF wins all seven magnitude-step scenarios and the family mean as well.

Table II also distinguishes a winner from a decisive winner. In the magnitude-step family, EKF's 1.45 mHz lead over RA-EKF is about one order of magnitude larger than the winner's mean within-scenario spread (0.143 mHz), so the aggregate advantage is meaningful. By contrast, the ramp, modulation, phase-jump, harmonic, ringdown, and out-of-band families all have family-level margins that are smaller than or comparable to the same spread. Those families should therefore be read as closely contested rather than dominated by a single method.

Trip-risk is even less discriminating in the clean standard cases. In 23 of the 34 scenarios, the minimum trip-risk is zero, often shared by 10 to 13 estimators in the ramp and magnitude-step families. In those cases, trip-risk is useful mainly as a pass/fail screen, while RMSE and peak error carry most of the ranking information.

Phase jumps behave differently. SOGI-FLL gives the lowest mean RMSE across the phase-jump family (174 mHz), but

ZCD gives the lowest mean trip-risk (0.0194 s). The protection and accuracy metrics therefore point to different designs. In the 60° phase-jump case, ZCD keeps trip-risk below the leading low-RMSE methods, but only with much larger error excursions. Even within the standard test set, RMSE and trip-risk are not interchangeable objectives.

B. *Composite IBR Stress Reorders the Ranking*

The broadest change appears when the disturbance becomes composite. The hardest benchmark scenarios, measured by median RMSE across estimators, are the IBR multi-event sequence (826 mHz), the IEEE and NERC 60° phase jumps (675 mHz and 637 mHz), and the noisy IBR ringdown scenarios (474–528 mHz). These are precisely the cases where the usual standard test intuition starts to fail.

The scale change is itself important. The easiest family in Table II, magnitude steps, stays below 1 mHz, whereas the IBR multi-event scenario reaches 826 mHz median RMSE across estimators and the noisy ringdown family sits near half a hertz. The benchmark therefore spans roughly three orders of magnitude in error severity: the median RMSE of the IBR multi-event case is about 1.55×10^3 times that of the single-tone case. That spread explains why rankings inferred from clean standard tests do not transfer directly to composite IBR stress.

The IBR multi-event sequence is the clearest example. PLL gives the lowest RMSE of 173.5 mHz. UKF trails by only 2.74 mHz on the family mean. EKF remains nearly tied in the scenario readout at 176.4 mHz while reducing trip-risk to 0.0874 s. IpDFT yields the lowest trip-risk (0.0439 s) but at the cost of a much larger peak error (2550 mHz). The ranking therefore depends on the deployment objective: minimizing average error, minimizing time above the trip threshold, or bounding transient severity.

The same pattern appears in the other IBR-specific families. EKF gives the lowest mean RMSE across the harmonic family (70.5 mHz), but TFT and Prony give the lowest mean trip-risk there (0.00947 s). In the ringdown and noise family, UKF gives the lowest mean RMSE (313 mHz), whereas ESPRIT gives the lowest mean trip-risk (0.0989 s) at a far higher computational cost. These results argue against a single global ranking. Scenario family is part of the problem definition, not a *post hoc* qualifier.

The scenario-level results make that split sharper. Among the 12 scenarios with a unique trip-risk winner, nine differ from the RMSE leader. Those disagreements concentrate in the 60° phase jumps and in the harmonic, ringdown, and multi-event IBR cases. The harmonic family is especially instructive: TFT wins two of the three individual harmonic scenarios, and RA-EKF wins the third. Yet EKF still achieves the best family-average RMSE because it stays consistently close to the scenario leaders across all three stress levels.

Composite stress also changes which metric carries the practical decision. In magnitude steps and clean ramps, many leading estimators already tie at zero trip-risk, so RMSE is the useful separator. In the multi-event, phase-jump, and ringdown families, RMSE, trip-risk, and peak excursion split across different estimators. That metric split, not just the change in rank order, is the core reason a single scalar ranking fails to describe protection-relevant behavior.

C. *Global Rankings Are Useful Only as Secondary Evidence*

Family-by-family reading is the main analysis, but the global summaries still help when they are interpreted carefully. The interpretation depends on the aggregation rule. Mean-rank summaries across the 34 benchmark scenarios favor TFT on RMSE (4.38) and peak error (4.32), with RA-EKF close behind on both (4.62 and 4.59). Simple averaging over the 34 benchmark scenarios instead favors RA-EKF and EKF, with mean RMSE values of 116 mHz and 123 mHz, followed by UKF at 131 mHz. That difference is informative rather than contradictory: mean rank rewards repeated near-front finishes, whereas raw averaging is driven more strongly by the hardest disturbance families.

Consistency statistics make that distinction concrete. EKF is the most frequent accuracy leader, with 10 outright RMSE wins, 19 RMSE top-three finishes, and 20 peak-error top-three finishes across the 34 benchmark scenarios. TFT wins only two RMSE scenarios, yet it reaches the RMSE top three in 16 scenarios and the peak-error top three in 17, which explains its strong mean-rank position. Trip-risk favors a different group again: IpDFT and ESPRIT enter the trip-risk top three in 25 and 24 scenarios, respectively. Global summaries are therefore best read as screening tools, not as substitutes for family-level evidence.

The MAE versus CPU Pareto front contains EKF, RA-EKF, SOGI-FLL, SOGI-PLL, and ZCD, indicating that neither accuracy nor cost alone determines the admissible set. As secondary evidence, the study also includes ANOVA and Kruskal-Wallis tests at the family level on RMSE, MAE, peak error, trip-risk, settling time, and CPU time; both tests reject the null hypothesis of equal family-level distributions for all six metrics (all $p < 10^{-16}$). These tests are coarse and do not replace the scenario-family analysis, but they support the claim that the observed ranking differences are not confined to one metric.

ZCD remains on the MAE versus CPU Pareto front because its cost is negligible, not because its transient accuracy remains competitive under phase jumps or multi-event stress. The practically relevant low-cost region is therefore narrower than the raw Pareto set suggests.

For deployment decisions, the recommendation remains scenario-dependent. SOGI-PLL is the strongest choice when ramp RMSE is the main objective, EKF is strongest on magnitude-step and out-of-band families, RA-EKF is strongest on modulation, and PLL is the lowest-RMSE option in the composite IBR event. None of those statements can be promoted to a universal recommendation, because the lowest trip-risk in the same family may belong to a different method. That conditional reading is stronger than a flat ranked list because it keeps both the disturbance family and the operating objective in view.

D. Cost Keeps Fast Embedded Methods in the Discussion

CPU cost changes the ranking again. Figure 2 gathers representative time-domain cases together with the global summaries. EKF and RA-EKF occupy the best global accuracy region with mean CPU costs of $0.813\,\mu$s and $1.61\,\mu$s per sample. SOGI-PLL and SOGI-FLL remain much cheaper ($0.0867\,\mu$s and $0.0348\,\mu$s) and therefore stay relevant for embedded protection even when they lose some accuracy. Window-based methods such as TFT and IpDFT improve global trip-risk relative to some loop methods, but they do so at about 3–4 μs per sample. ESPRIT reaches the trip-risk top three almost as often as IpDFT, but at $2838\,\mu$s per sample instead of $3.68\,\mu$s. Koopman and ESPRIT therefore occupy a substantially higher-cost region, with mean costs of $390\,\mu$s and $2838\,\mu$s, without a matching global RMSE gain. The time-domain panels illustrate the same point: the estimator that performs best in the ramp case is not the one that best manages the multi-event or ringdown case.

Taken together, the results suggest two practically relevant operating regions. EKF and RA-EKF define the low-error middle-cost region: they are not the cheapest estimators, but they remain competitive across the widest range of families and give the strongest aggregate balance once composite IBR stress is included. SOGI-PLL and SOGI-FLL define the low-cost embedded region: they keep sub-microsecond cost while remaining credible on ramps and selected phase-jump conditions. Window-based methods such as TFT and IpDFT remain useful when lower trip-risk justifies extra compute, but the more expensive reduced-order methods do not show a broad enough error advantage in this benchmark to justify routine deployment.

IV. Conclusion

This paper examined frequency estimation for low-inertia IBR grids through a broad Monte Carlo benchmark rather than a small set of isolated tests. The main finding is clear: estimator rankings move with the disturbance family. SOGI-PLL is strongest on ramps, EKF on magnitude steps and out-of-band interference, RA-EKF on modulation, and the IBR multi-event case splits RMSE and trip-risk winners between PLL and IpDFT. The hardest scenarios are the IBR multi-event sequence, the 60° phase jumps, and the noisy ringdown variants, which expose error patterns that do not appear in cleaner standard tests. More broadly, nine different estimators win at least one RMSE scenario, and among the 12 scenarios with a unique trip-risk winner, nine have a trip-risk winner that differs from the RMSE winner.

For practice, estimator choice should be tied to the protection objective. Fast loop-based methods remain attractive under tight computational budgets. EKF and RA-EKF offer the best overall balance across disturbance families in this benchmark. Selected window-based methods still matter when lower trip-risk under composite stress justifies higher cost.

The study is limited to the disturbance set and relative computational-cost measurements presented here. Hardware-in-the-loop validation and wider coverage of converter control interactions are the natural next steps.

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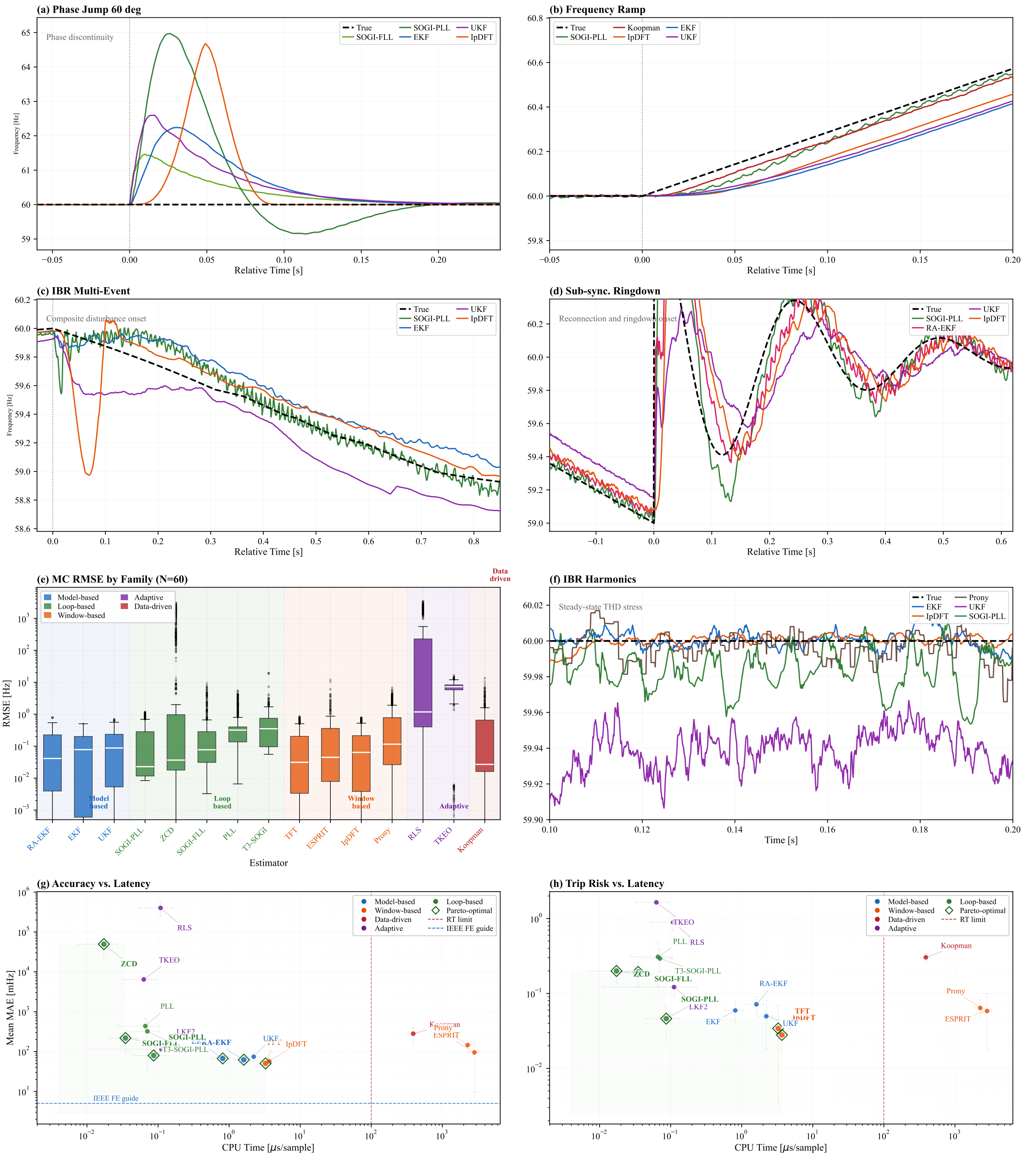

Fig. 2. Representative benchmark views from the study. Panels (a)–(d) and (f) show single-scenario responses, each summarized over the same 60 Monte Carlo runs used throughout the paper; panels (e), (g), and (h) summarize all 34 benchmark scenarios. (a) Phase-jump response. (b) Ramp response. (c) IBR multi-event response. (d) Ringdown response. (e) RMSE distributions by estimator family. (f) Harmonic-stress response. (g) Global mean MAE versus CPU time. (h) Global mean trip-risk versus CPU time. Together the panels show why the benchmark should be interpreted by disturbance family and operating objective rather than by a single scalar score.